Institut Universitaire des Sciences

Faculté des Sciences et Technologies (FST)

TD_3 – Réseaux 1

Rapport du travail de Laboratoire #3

• **Présenté par :** Kenley CHERY

• **Niveau :** 3ème année de Licence

• **Professeur :** Ismael SAINT AMOUR

• **G-mail :** kenley.chery@ius.education

Date : 3 Novembre 2026

L’objectif de ce TD est de :

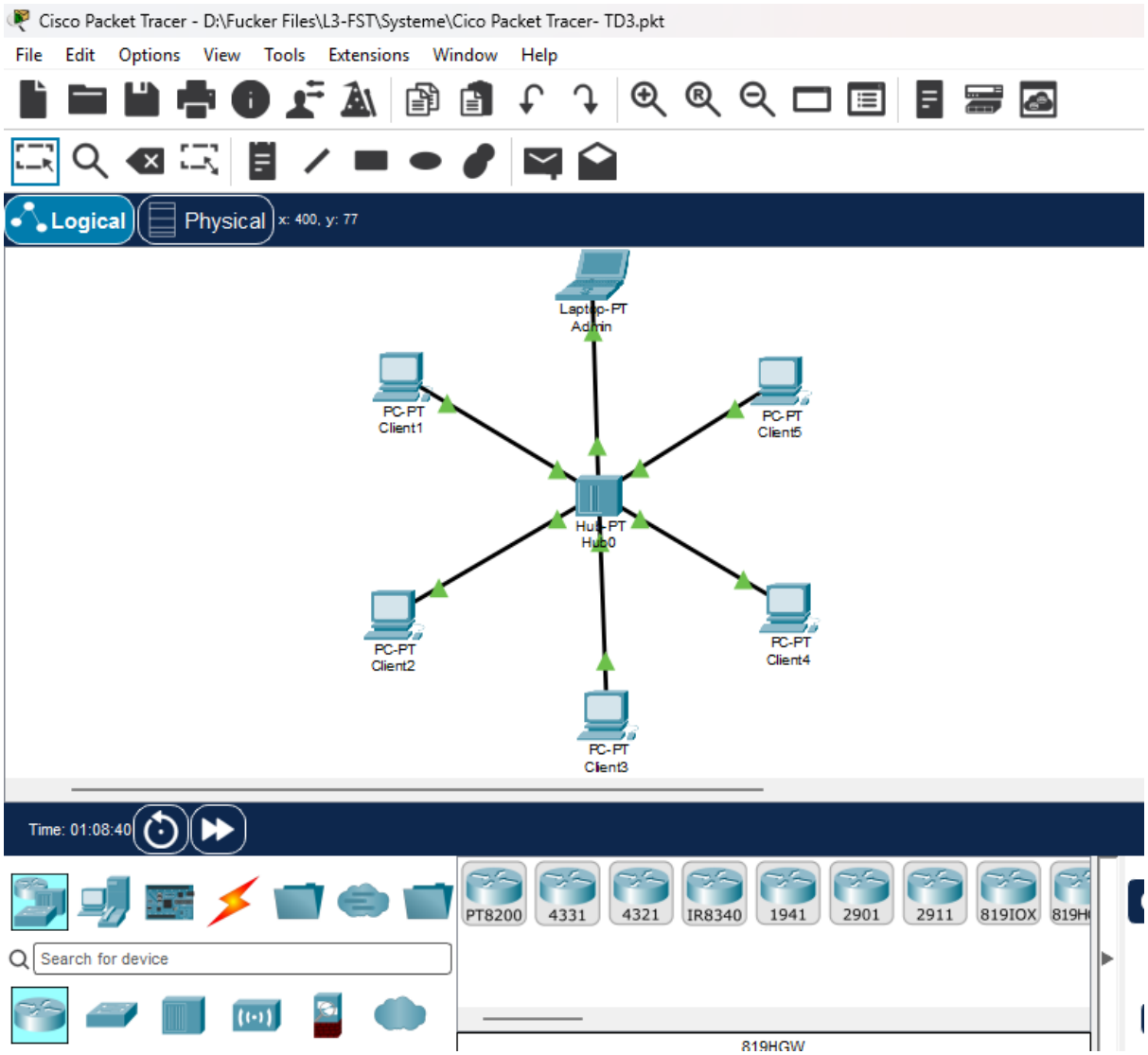
1. Savoir attribuer des adresses IP valides aux machines.
2. Comprendre l’adressage IPv4 et IPv6.
3. Vérifier la connectivité avec les commandes ping

Démarche suivie :

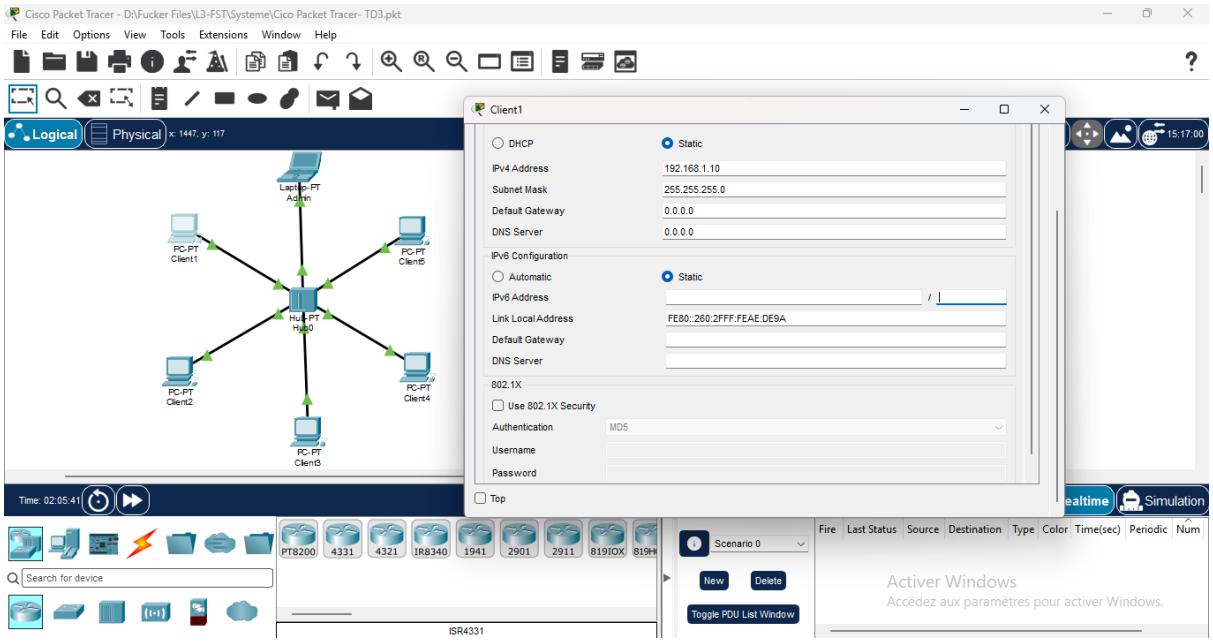
1- Construction de la topologie sur Cisco Packet Tracer
2- Configuration réseau IPv4 et vérification
3- Configuration réseau IPv6 et vérification

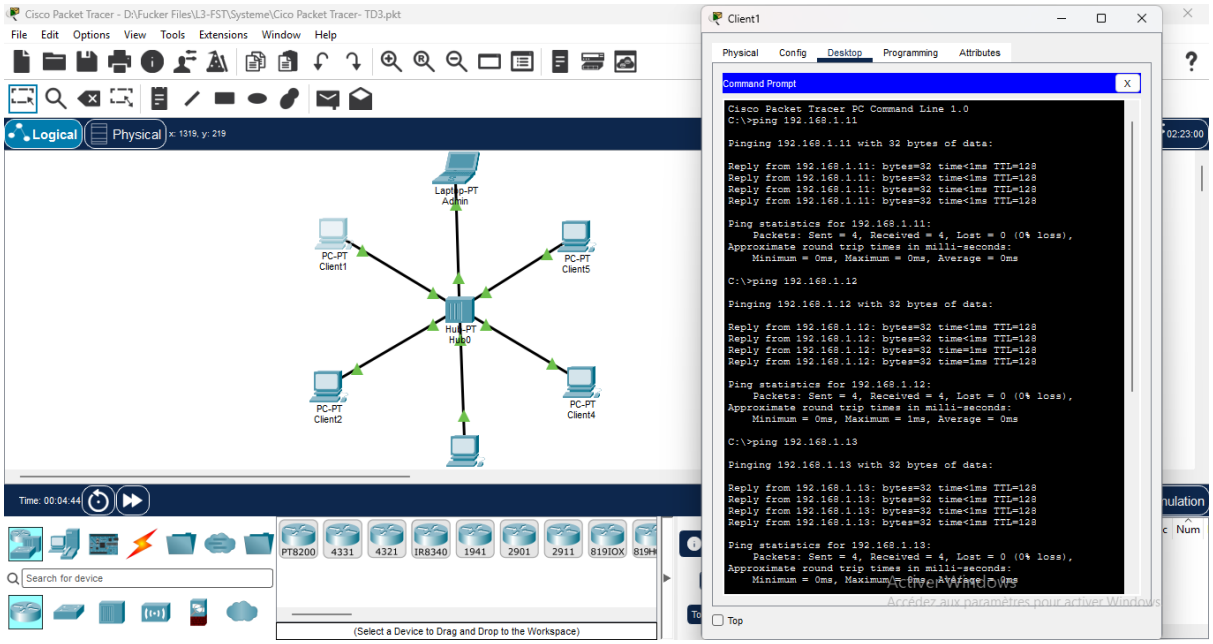
Étape 1 –

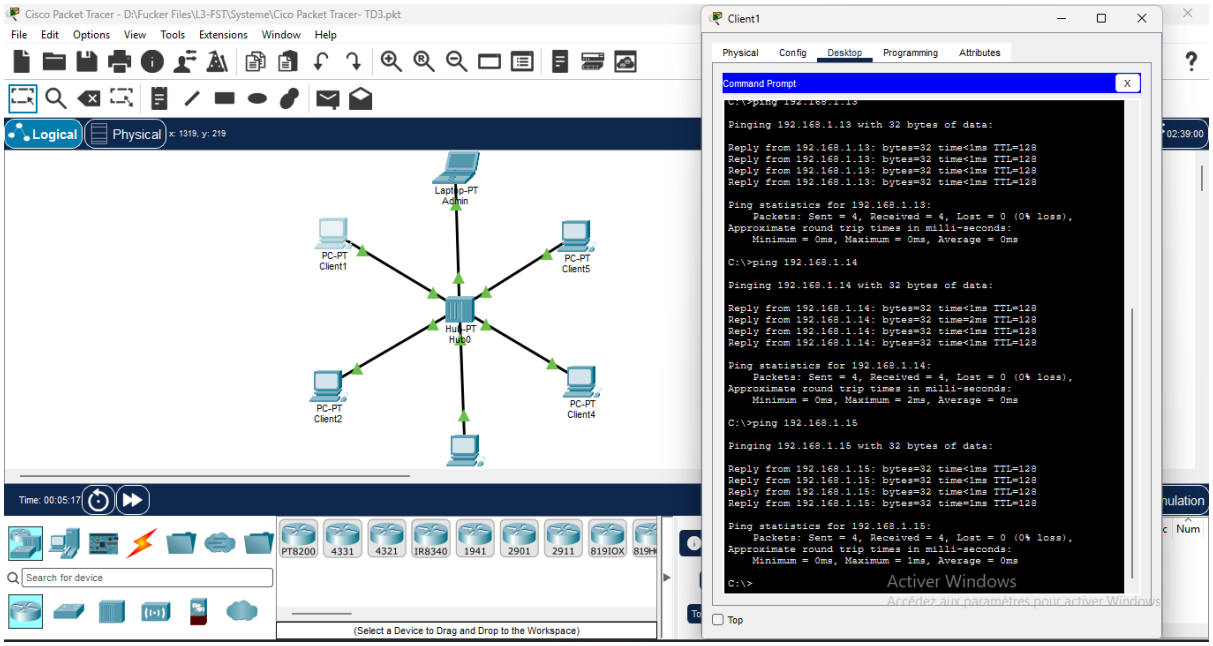
- Nous avons choisi la Topologie en Étoile de : 5 pc, 1 laptop et un Hub.
- Nous allons la configurer en IPv4 et vérifier les connectivités.



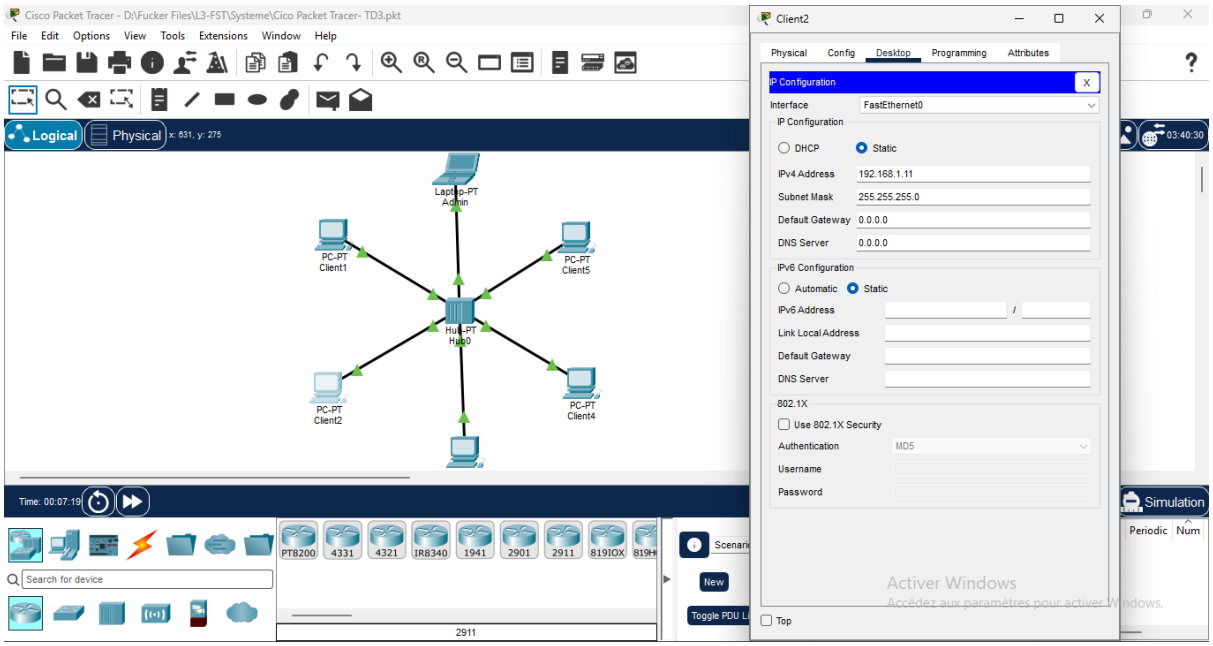
1- Configuration et verification du **Client1** en IPv4 :

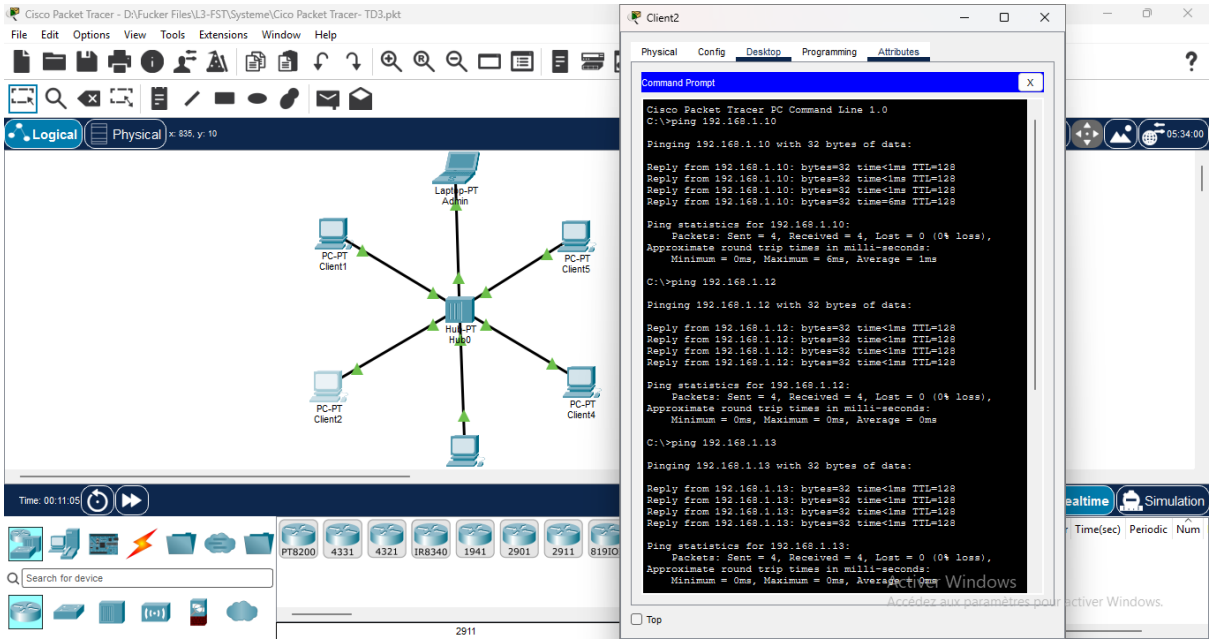

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical
Client1
DHCP
Static
IPv4 Address 192.168.1.10
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::260:2FFF:FEAE:DE9A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 02:05:41
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num
Search for device
PT8200 4331 4321 IR8340 1941 2901 2911 819IOX
ISR4331
Activer Windows
Accédez aux paramètres pour activer Windows.

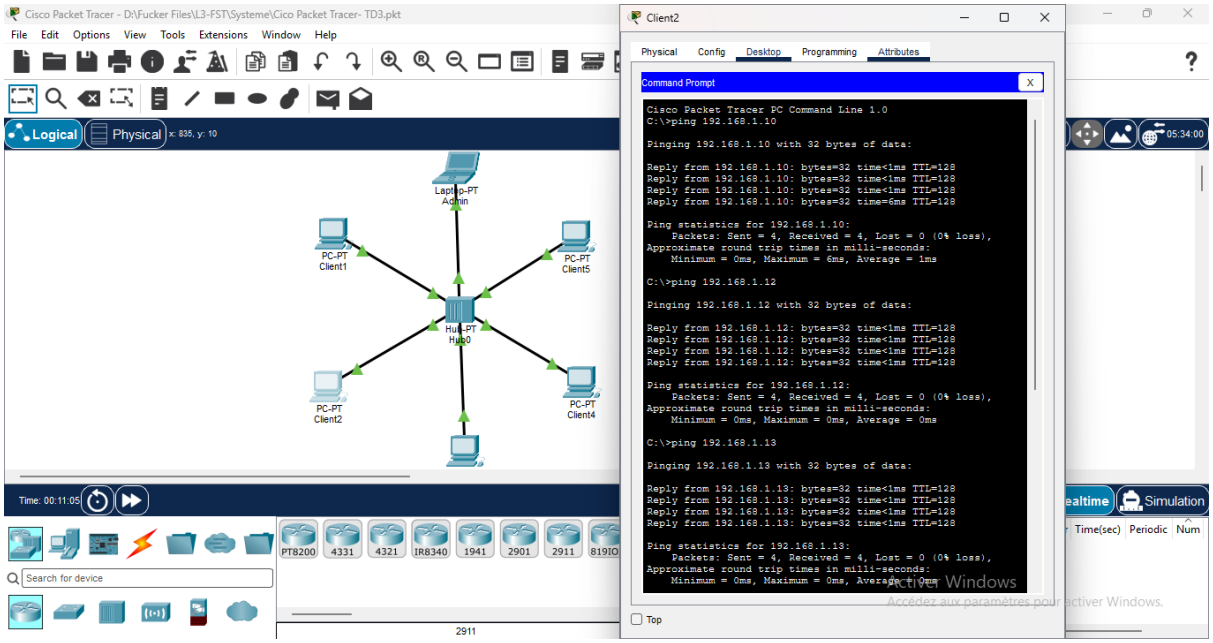

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical
Client1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
Time: 00:04:44
Search for device
PT8200 4331 4321 IR8340 1941 2901 2911 819IOX
(Select a Device to Drag and Drop to the Workspace)
Activer Windows
Accédez aux paramètres pour activer Windows.



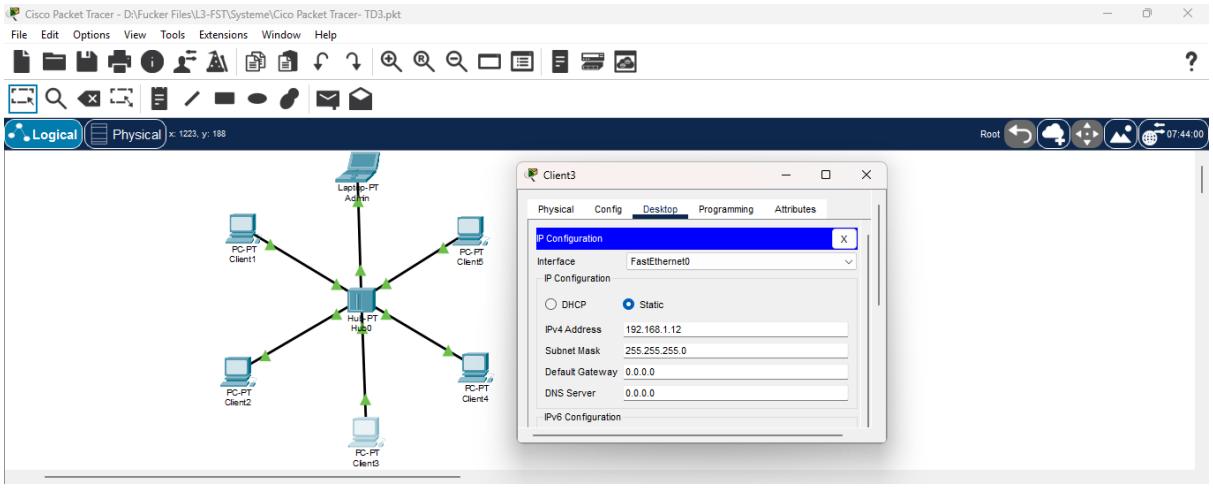
2- Configuration et verification du **Client2** en IPv4 :

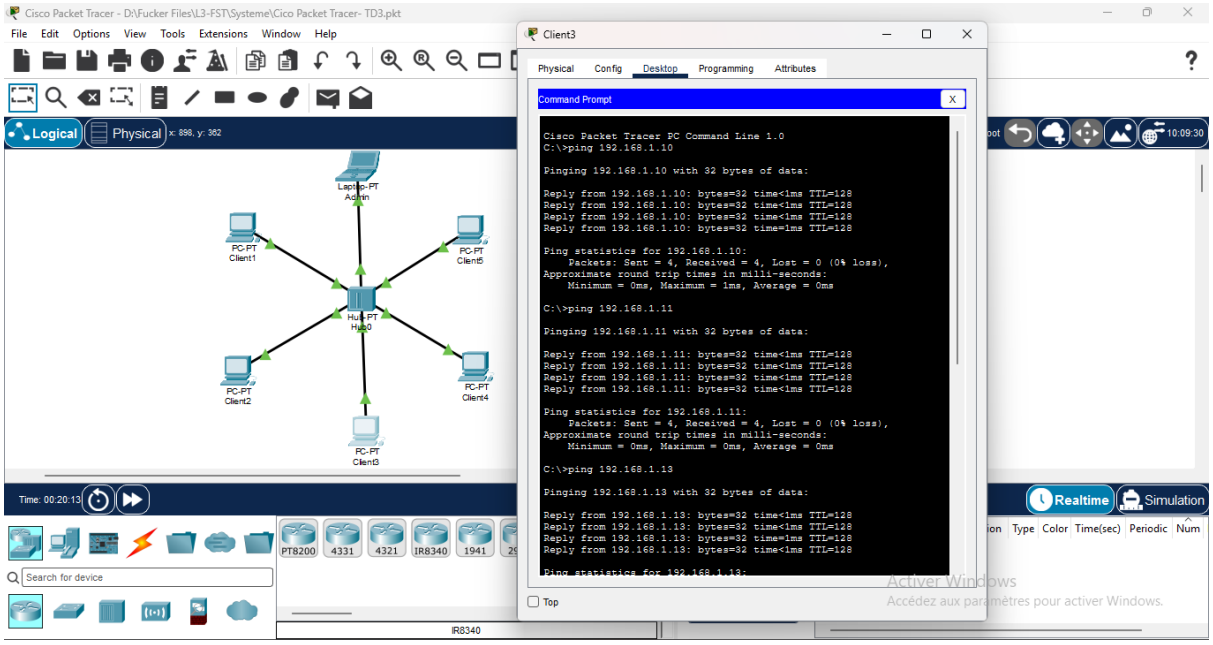


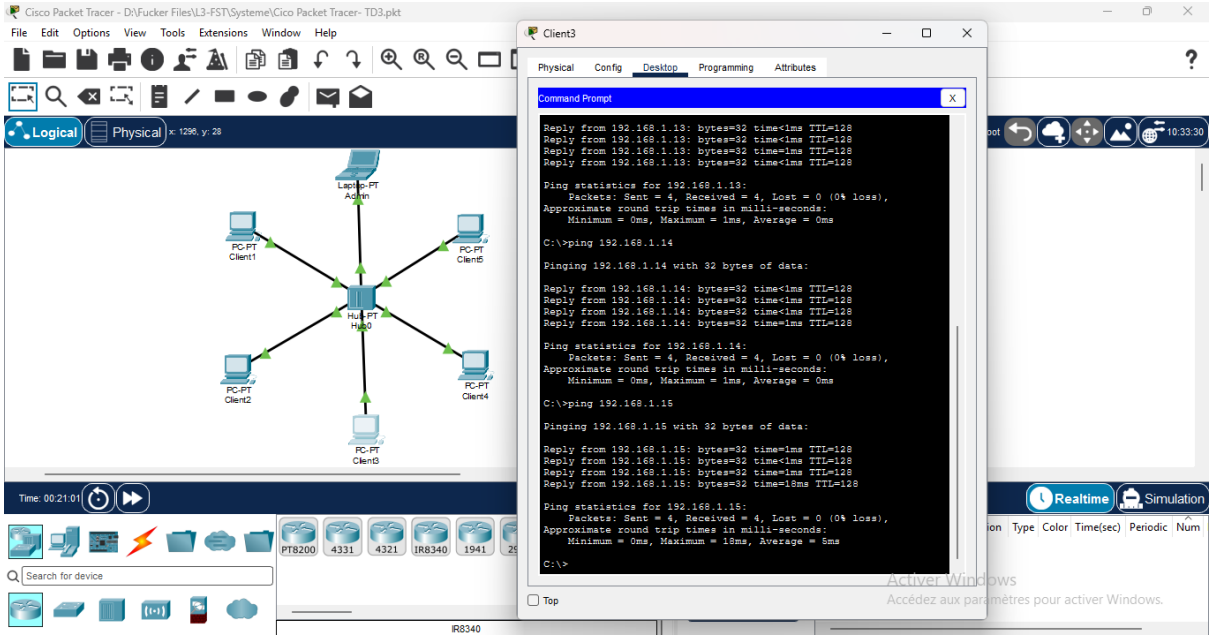




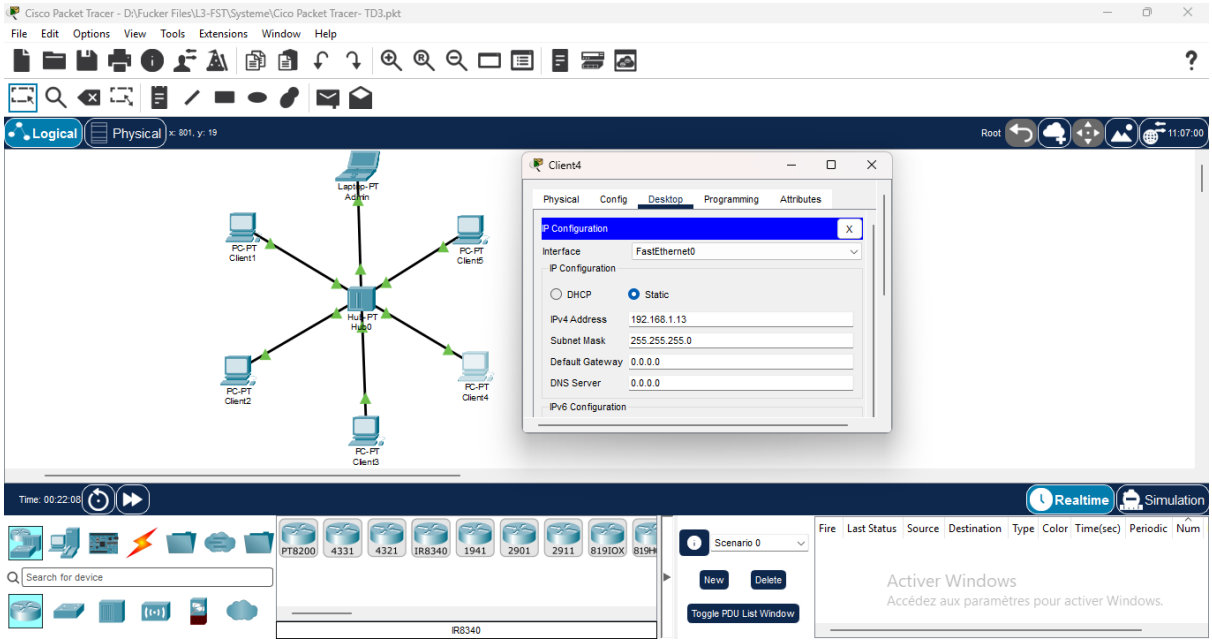
3- Configuration et verification du **Client3** en IPv4 :

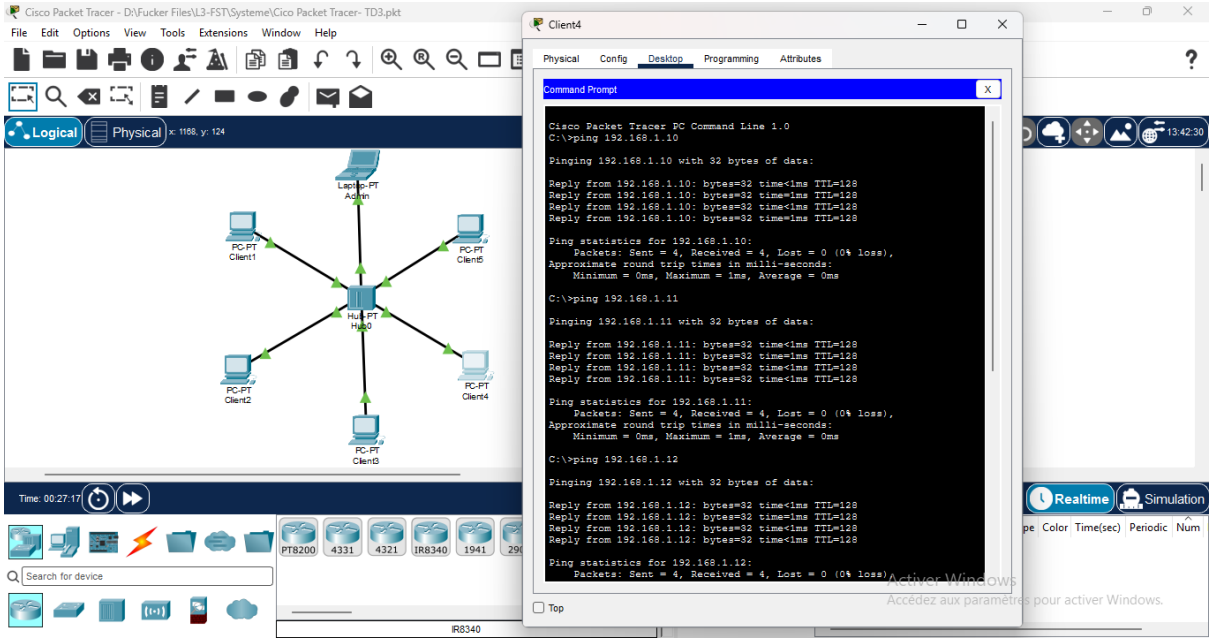

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical
Root
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client3
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 192.168.1.12
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration

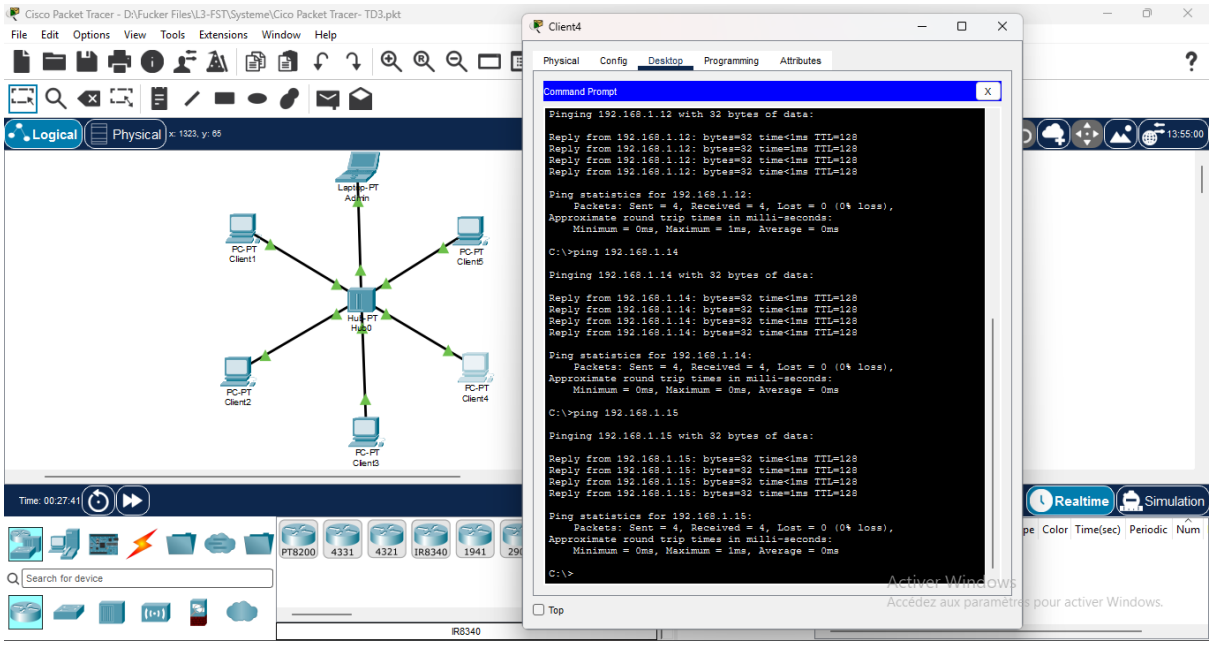

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical
Client3
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Realtime Simulation
Activer Windows


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical
Client3
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time=1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.14
Pinging 192.168.1.14 with 32 bytes of data:
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time<1ms TTL=128
Reply from 192.168.1.14: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.14:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.15
Pinging 192.168.1.15 with 32 bytes of data:
Reply from 192.168.1.15: bytes=32 time=1ms TTL=128
Reply from 192.168.1.15: bytes=32 time<1ms TTL=128
Reply from 192.168.1.15: bytes=32 time=1ms TTL=128
Reply from 192.168.1.15: bytes=32 time=18ms TTL=128
Ping statistics for 192.168.1.15:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 18ms, Average = 5ms
C:\>
Realtime Simulation
Activer Windows

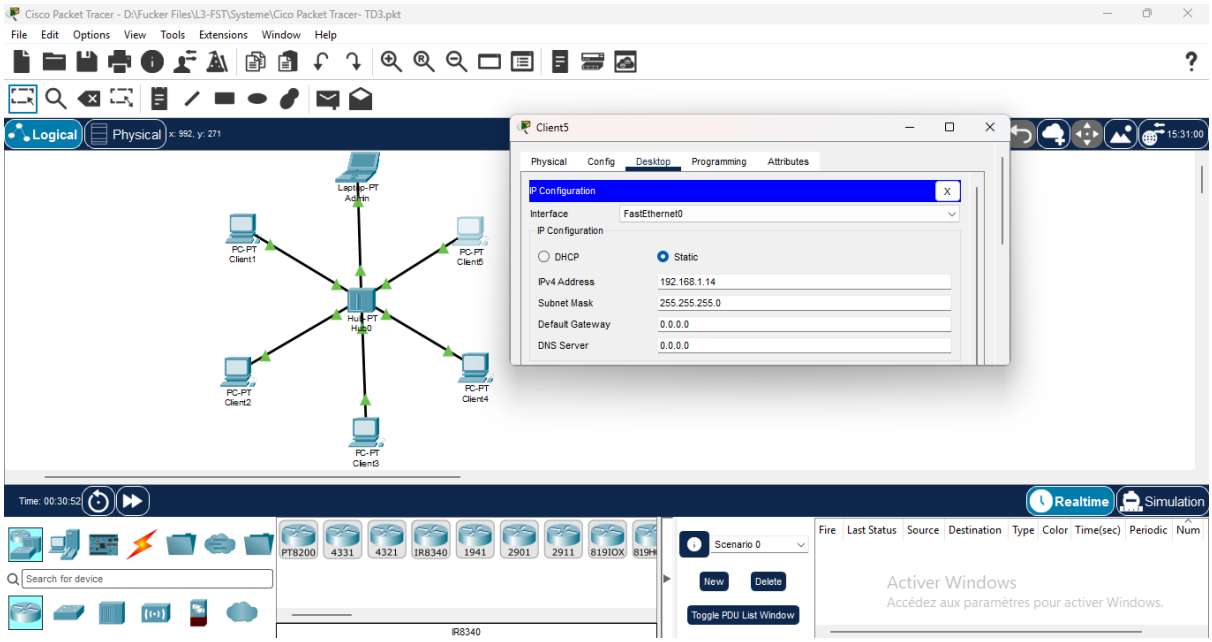
4- Configuration et verification du **Client4** en IPv4 :

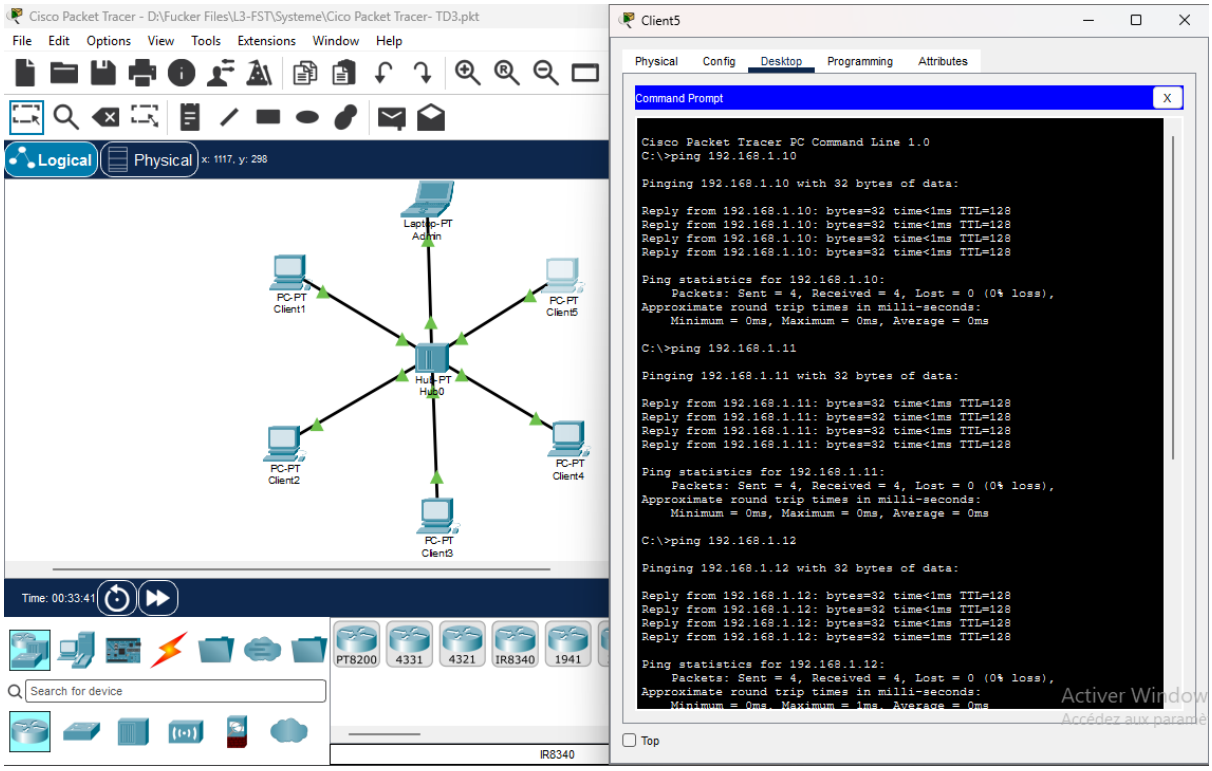


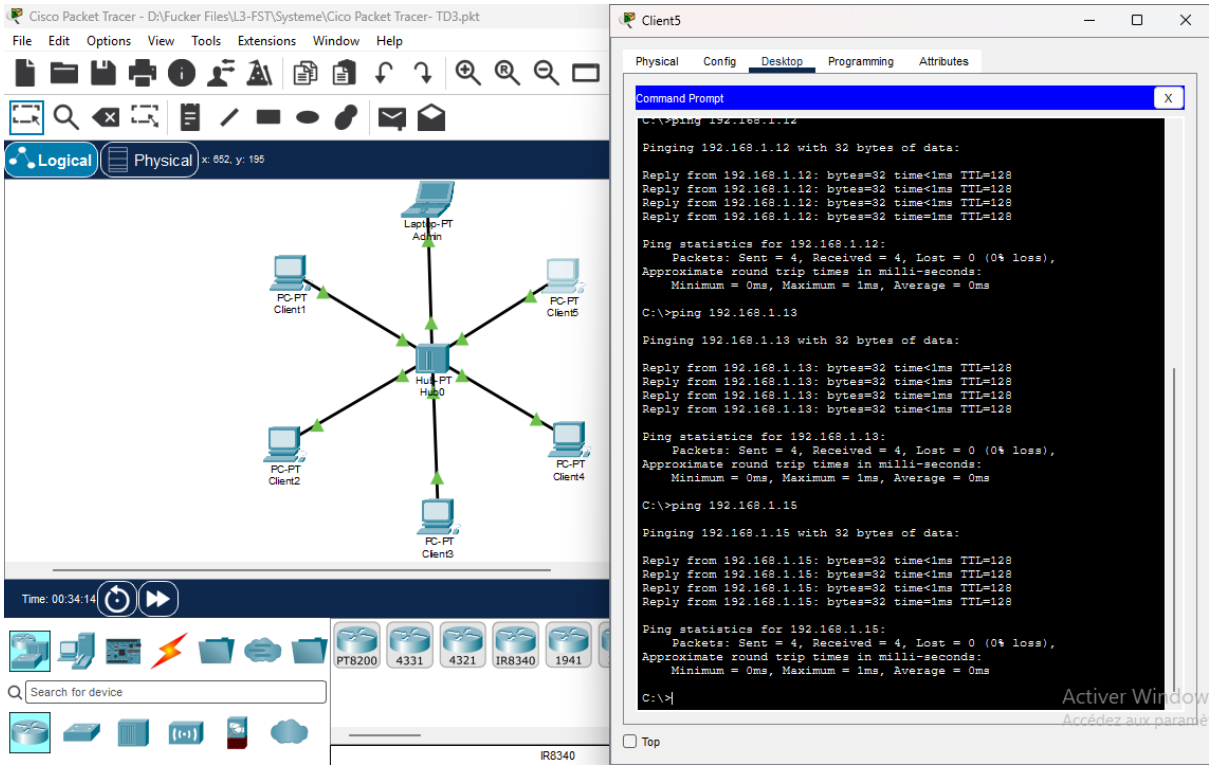




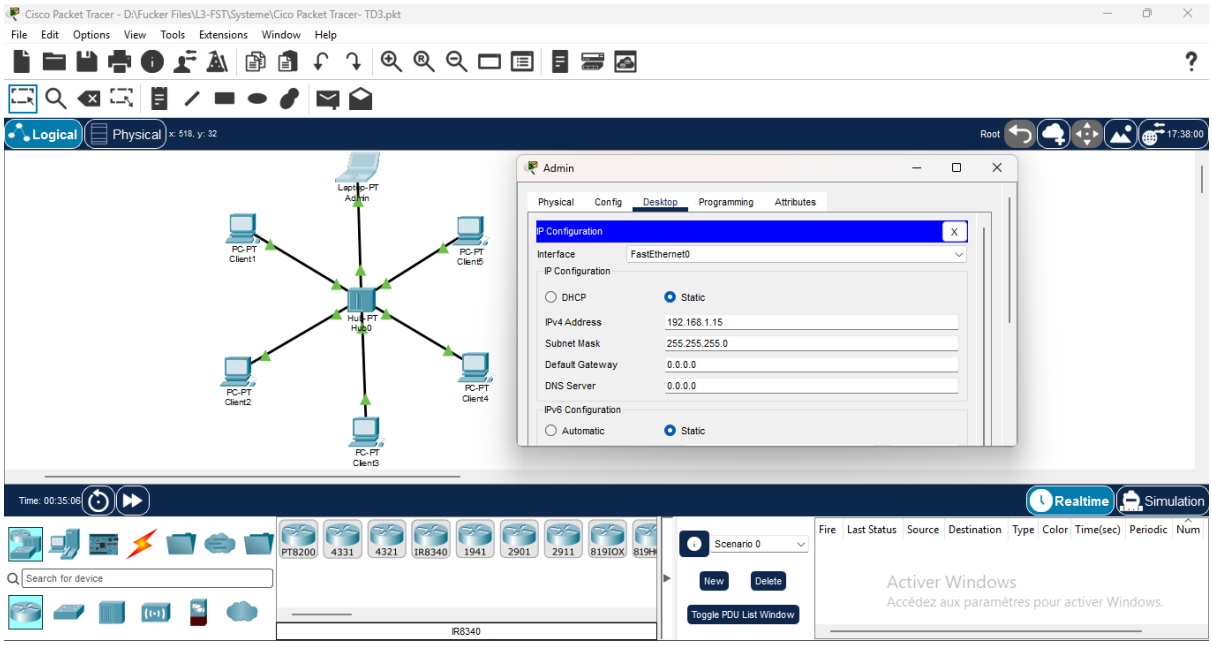
5- Configuration et verification du **Client5** en IPv4 :

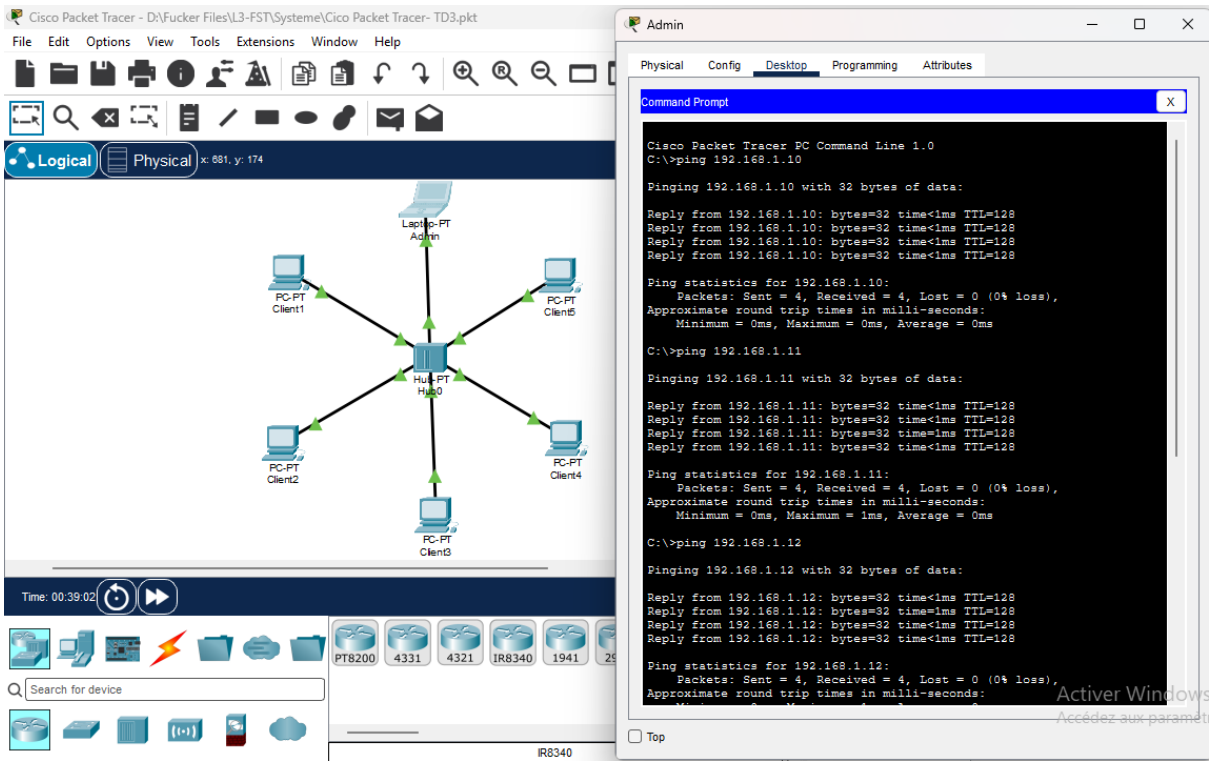


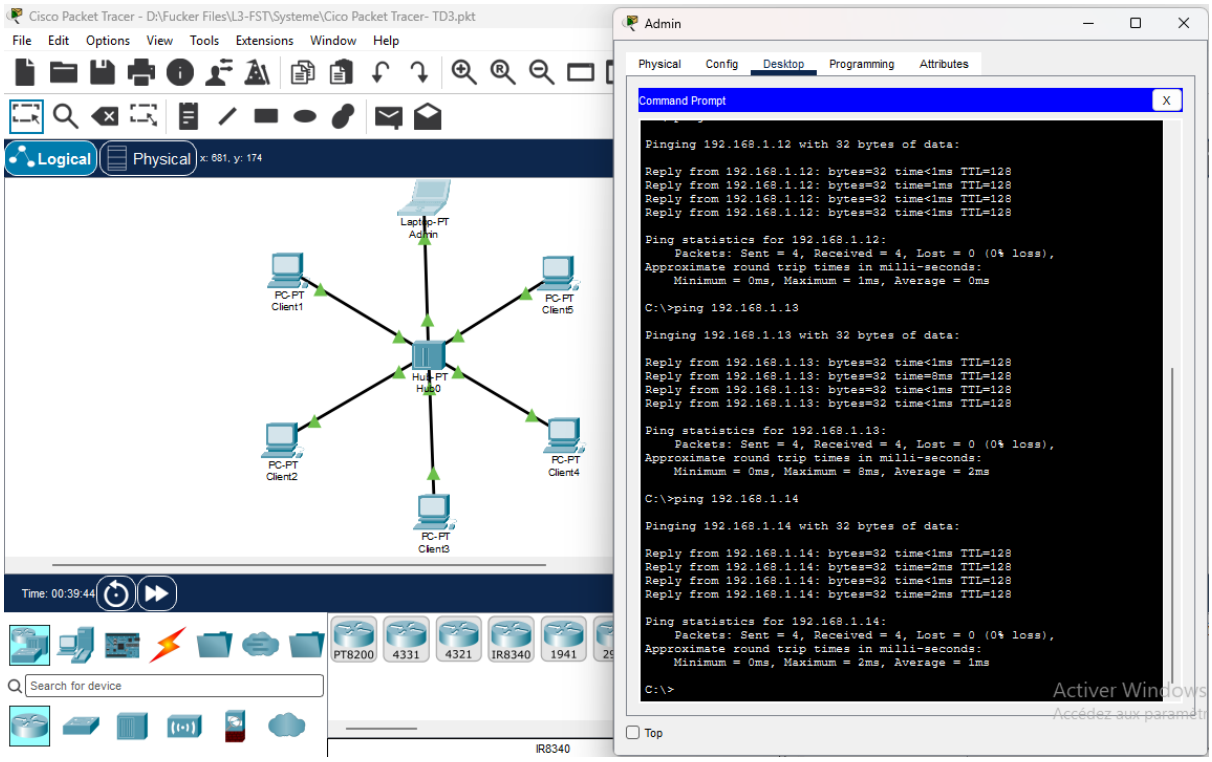




6- Configuration et verification du **Laptop Admin** en IPv4 :

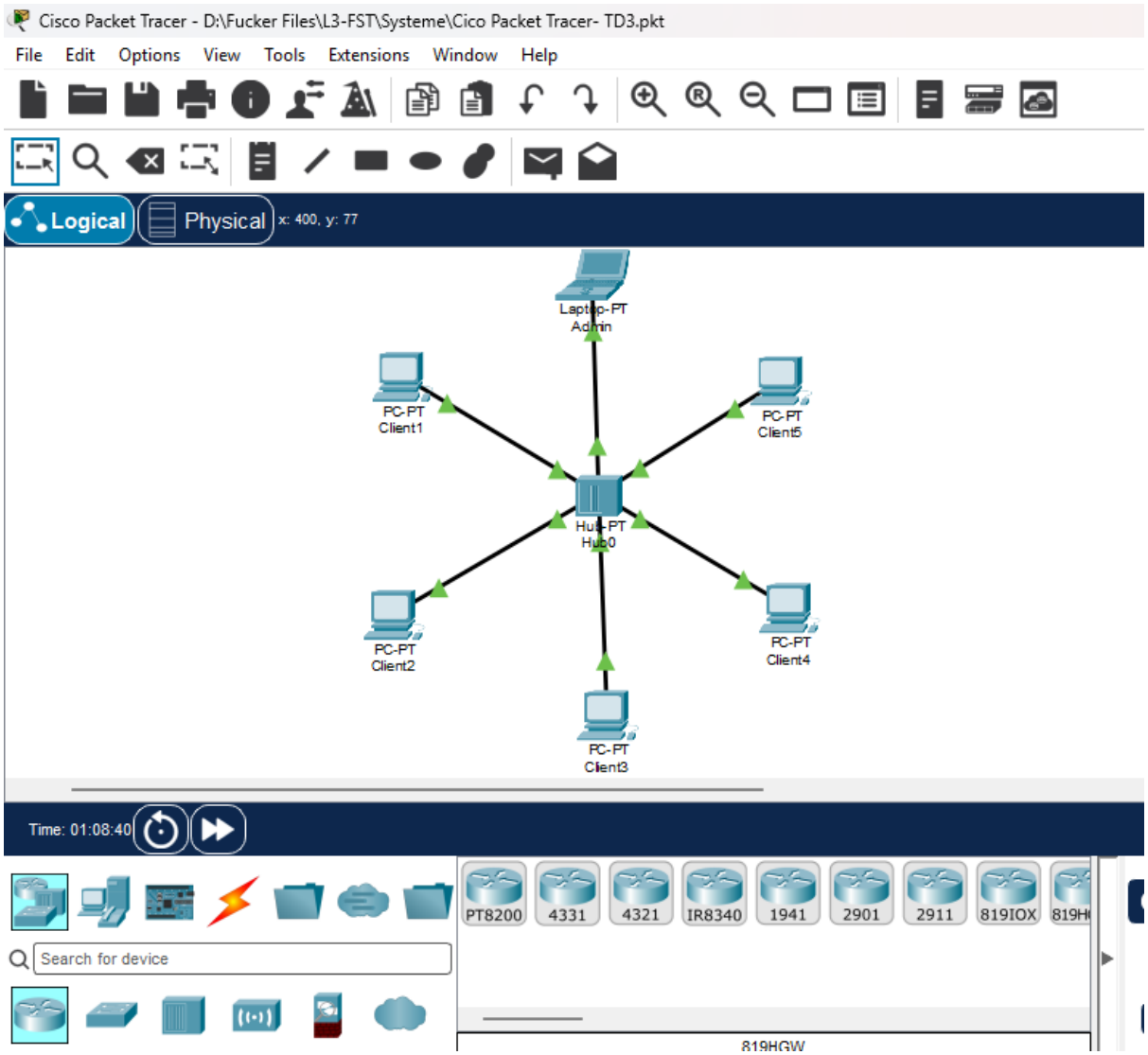

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Admin
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
DHCP
Static
IPv4 Address 192.168.1.15
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:35:06
Realtime
Simulation
Search for device
IR8340
Scenario 0
New
Delete
Toggle PDU List Window
Activer Windows


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Admin
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Top
Time: 00:39:02
Search for device
IR8340
Activer Windows

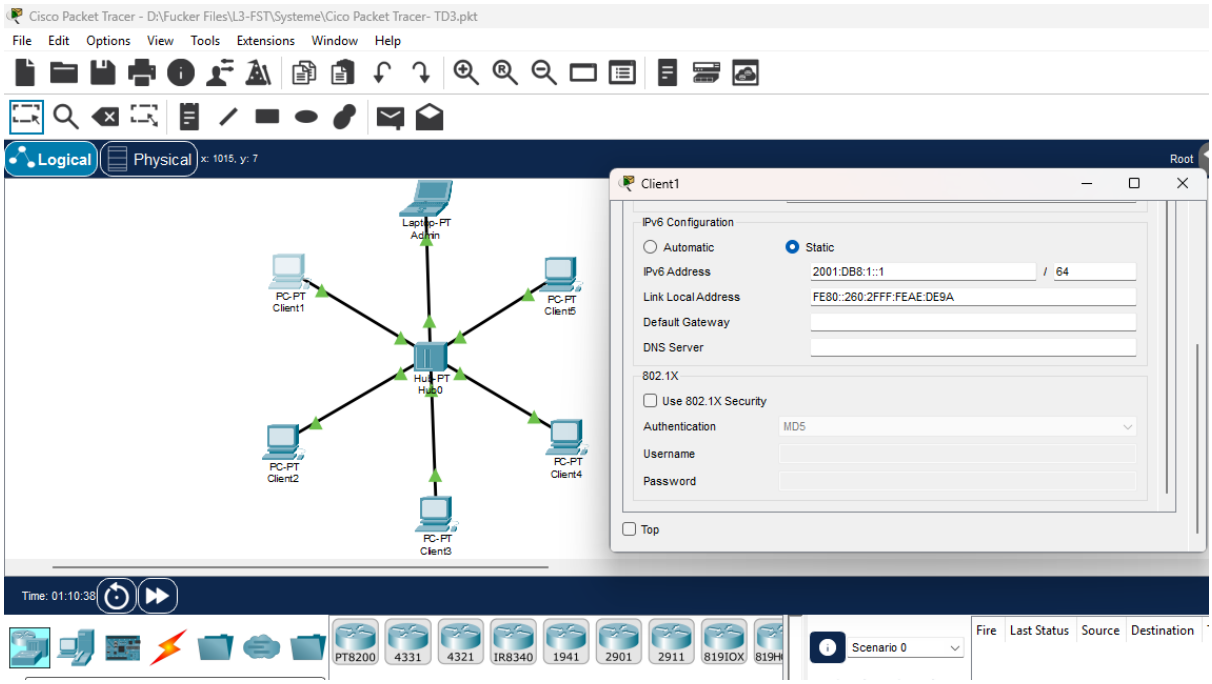


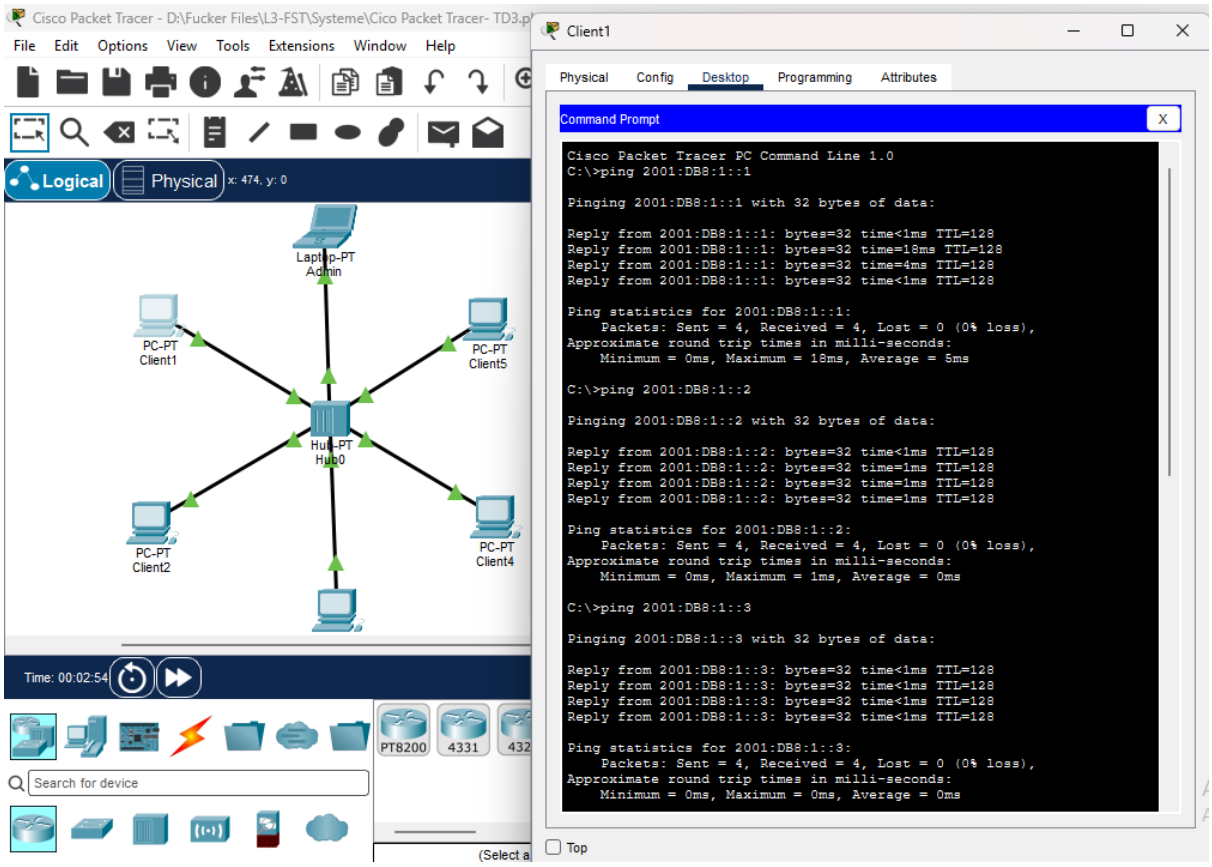
Étape 2 –

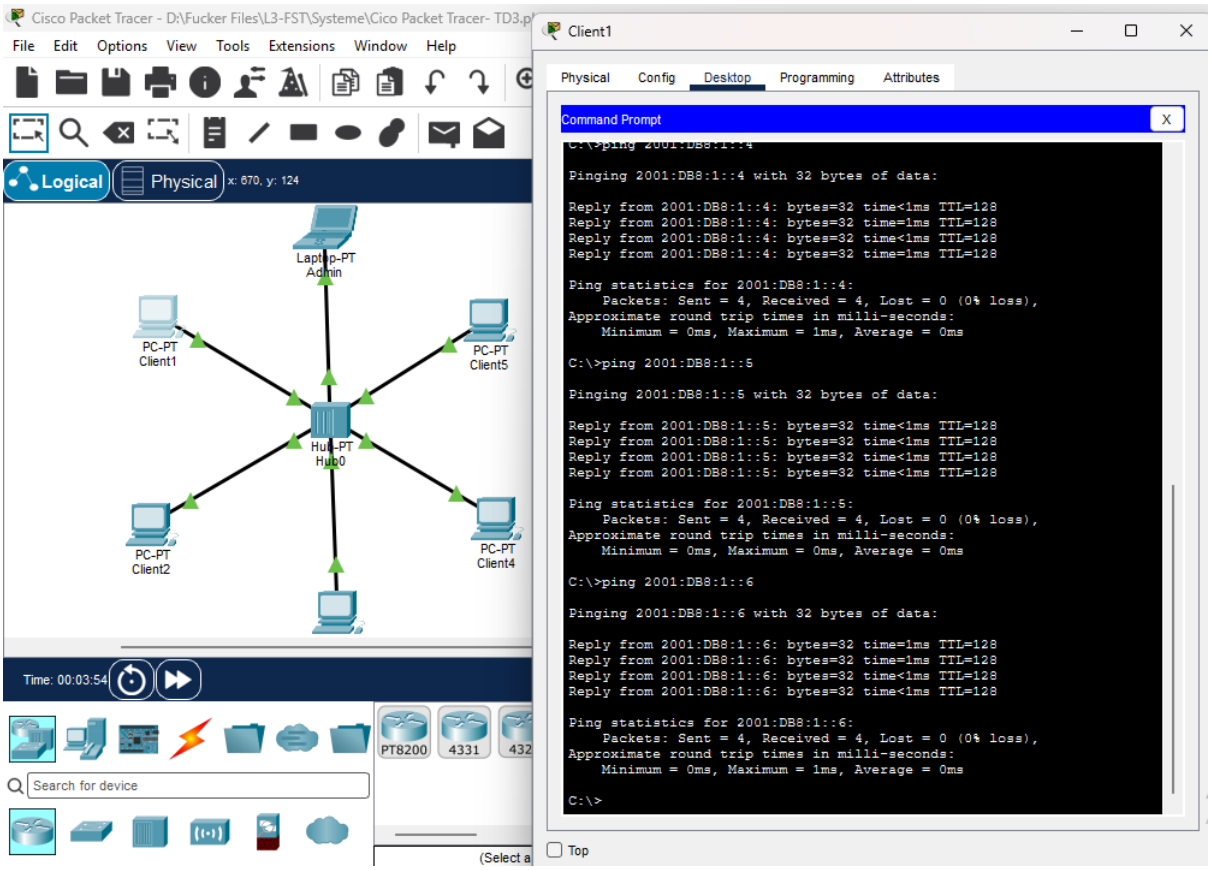
- On a fait le même choix du Topologie en Étoile de : 5 pc, 1 laptop et un Hub. Pour la configuration et la verification des connectivités en IPv6.



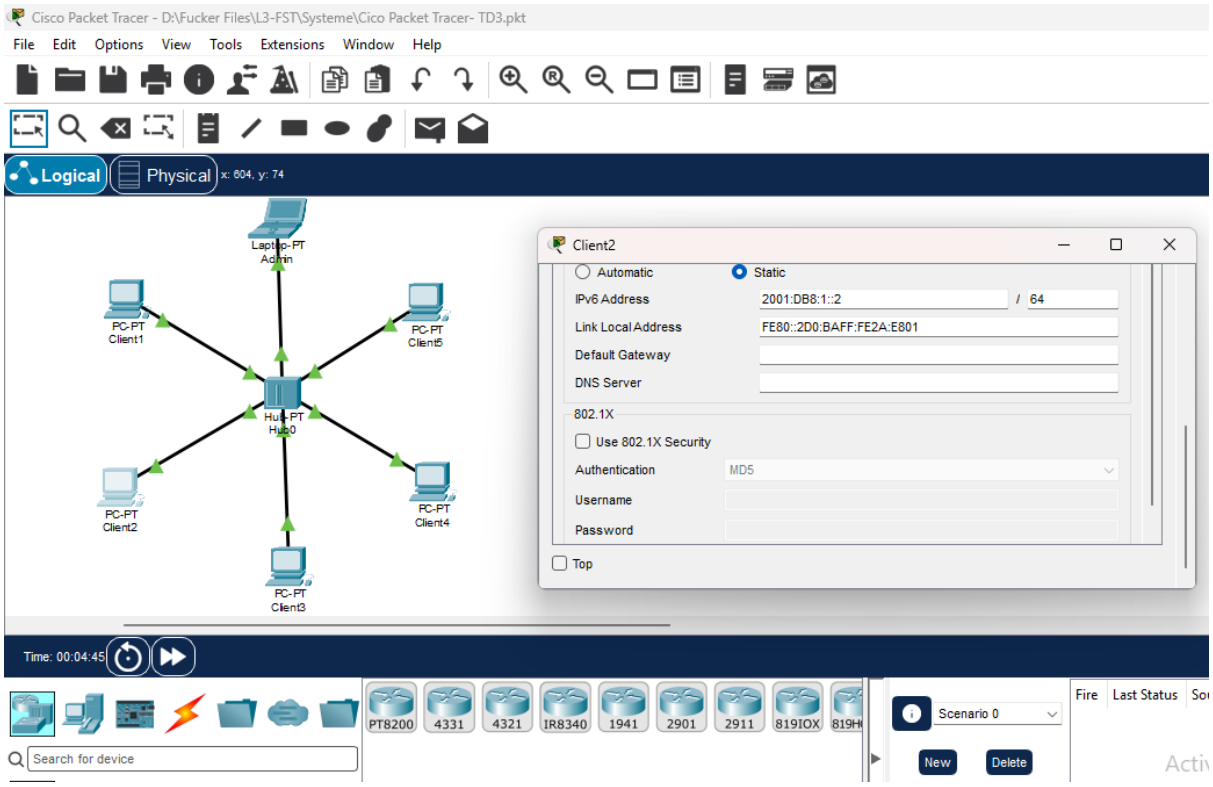
1- Configuration et verification du **Client1** en IPv6 :

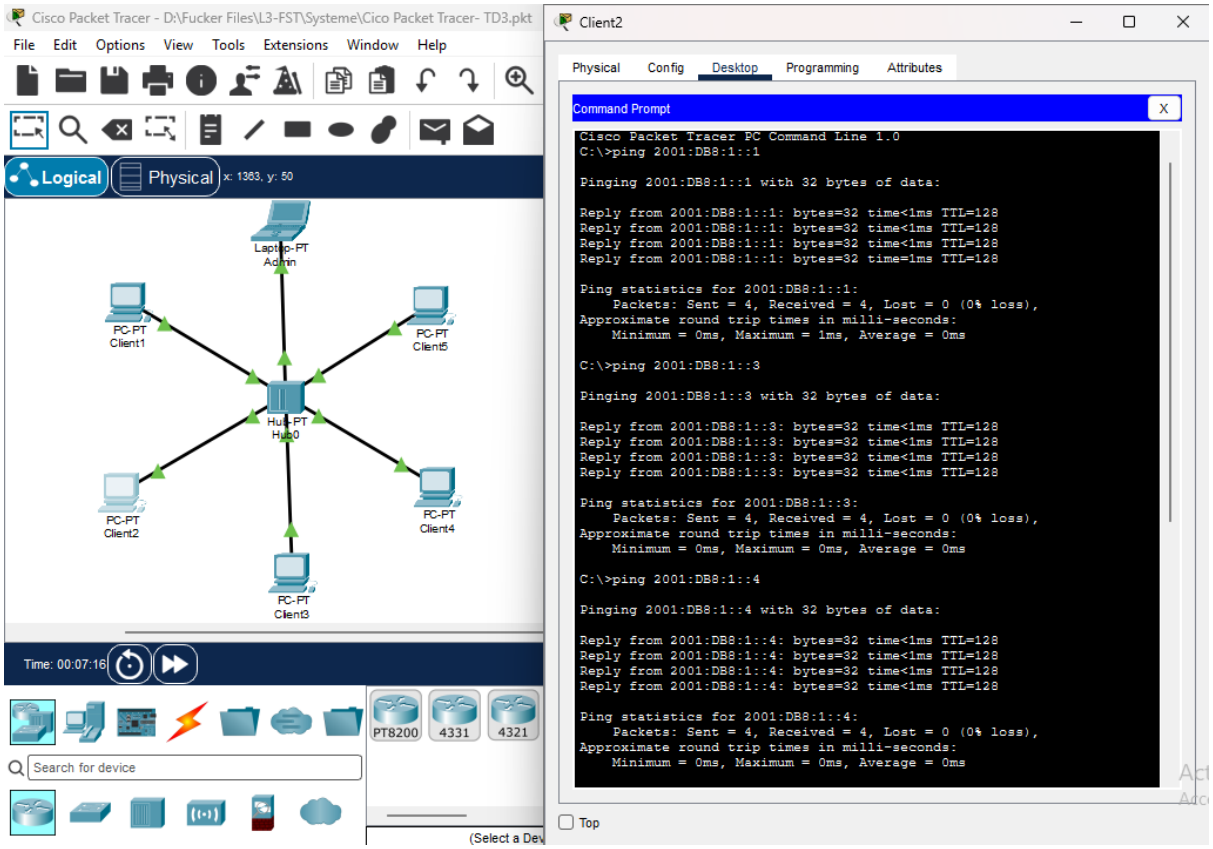

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1015, y: 7
Root
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client1
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::1
/ 64
Link Local Address
FE80::260:2FFF:FEAE:DE9A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Time: 01:10:38
PT8200
4331
4321
IR8340
1941
2901
2911
819IOX
Scenario 0
Fire
Last Status
Source
Destination

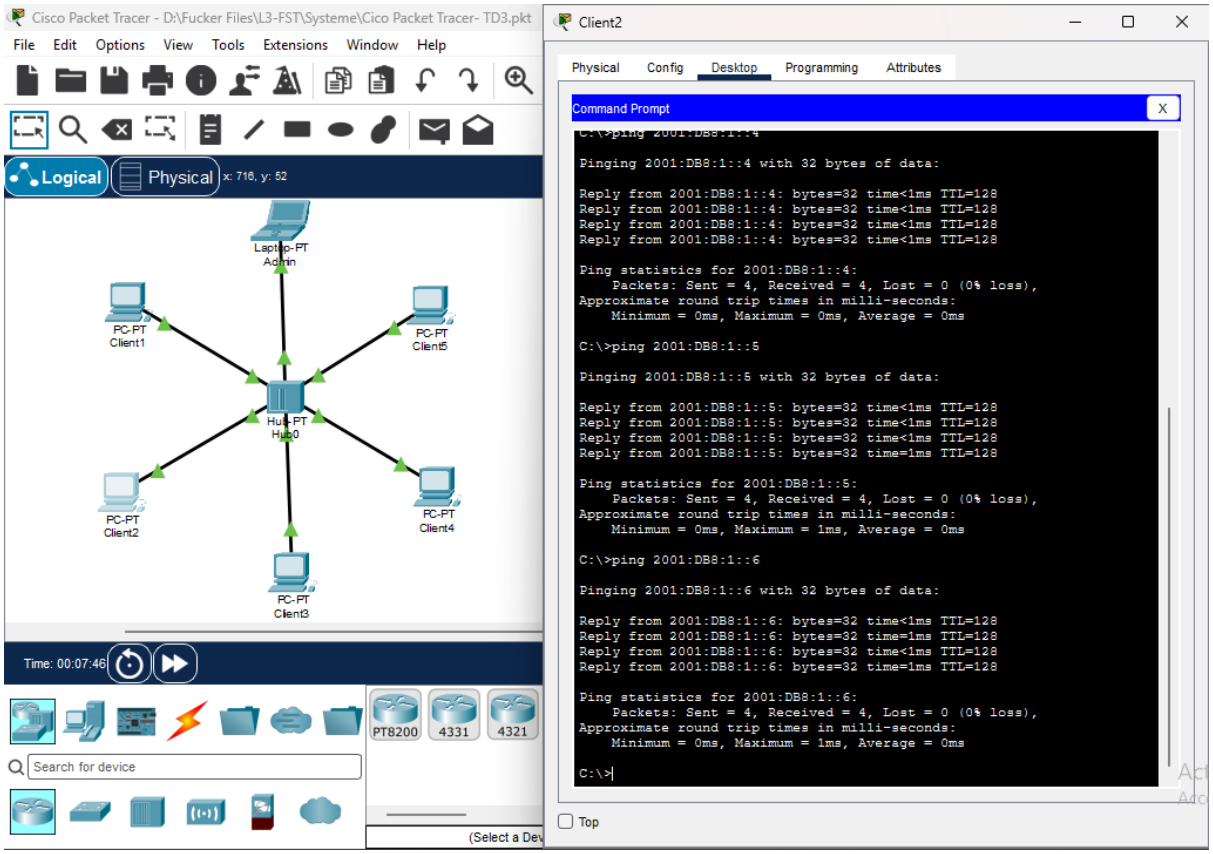

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.p
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 474, y: 0
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
Time: 00:02:54
PT8200
4331
Search for device
Client1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=18ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 18ms, Average = 5ms
C:\>ping 2001:DB8:1::2
Pinging 2001:DB8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::3
Pinging 2001:DB8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top



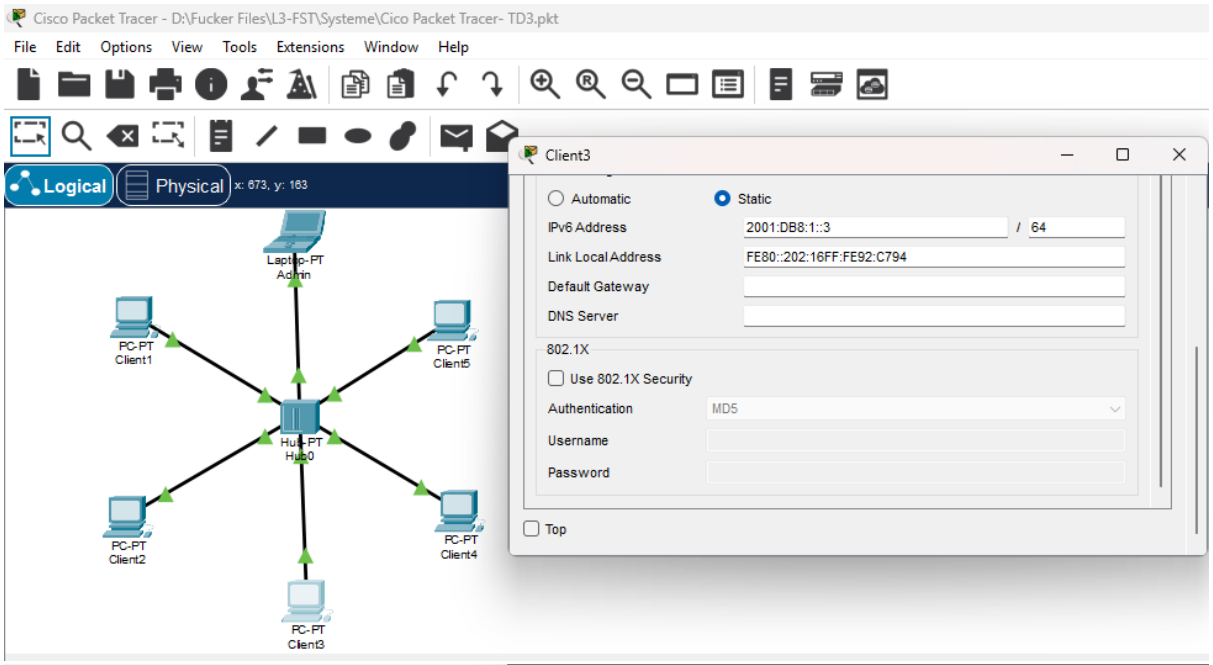
2- Configuration et vérification du **Client2** en IPv6 :

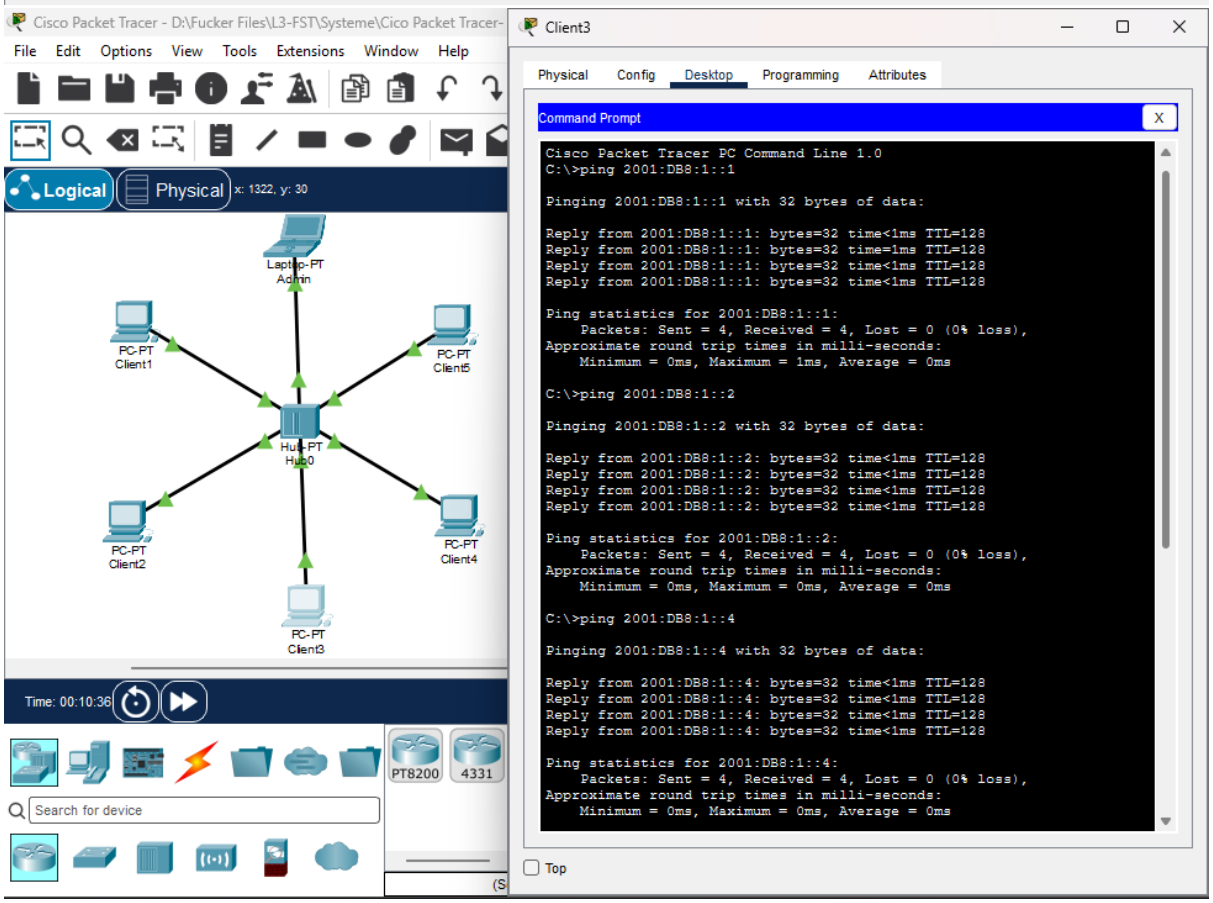

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 604, y: 74
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client2
Automatic
Static
IPv6 Address
2001:DB8:1::2
/ 64
Link Local Address
FE80::2D0:BAFF:FE2A:E801
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Time: 00:04:45
PT8200
4331
4321
IR8340
1941
2901
2911
819IOX
Search for device
Scenario 0
New
Delete
Fire
Last Status

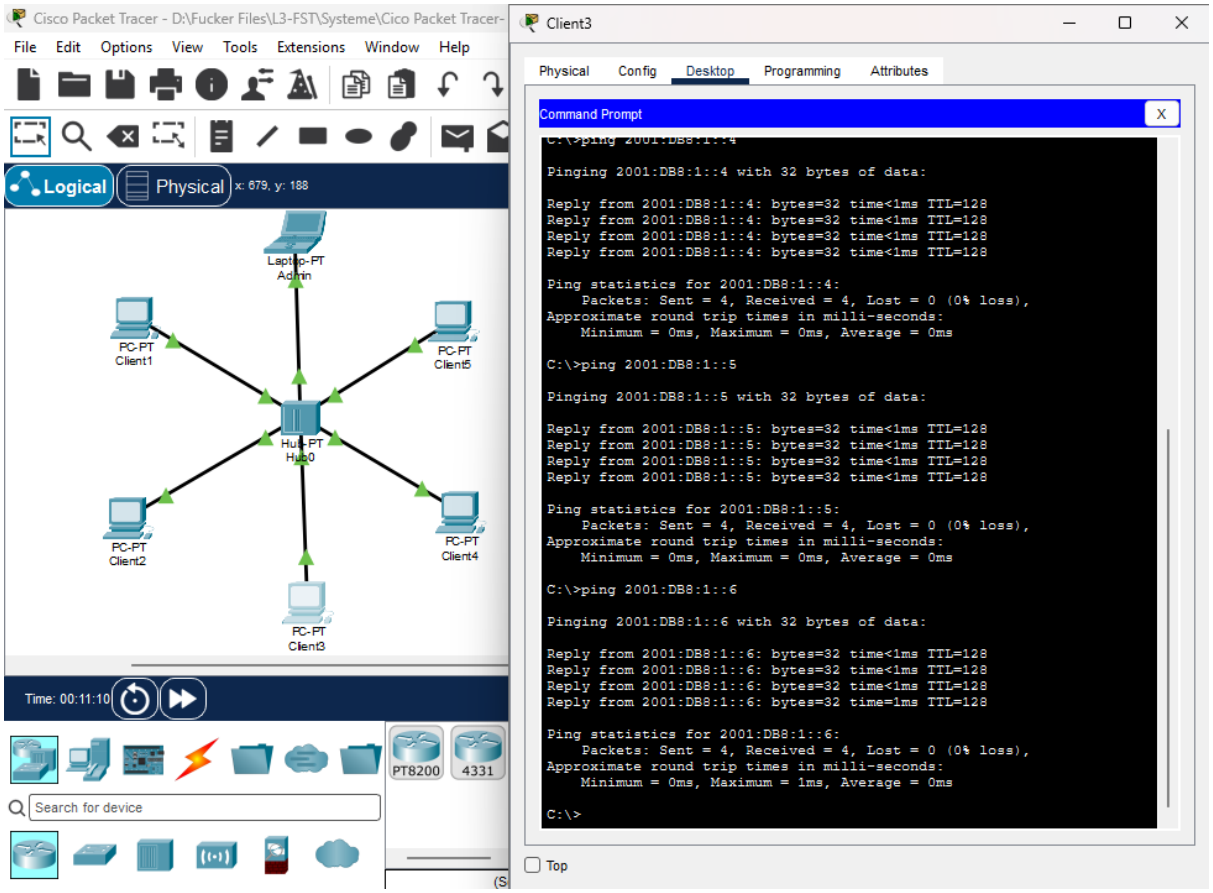

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1383, y: 50
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:07:16
PT8200
4331
4321
Search for device
Client2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::3
Pinging 2001:DB8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:DB8:1::4
Pinging 2001:DB8:1::4 with 32 bytes of data:
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top



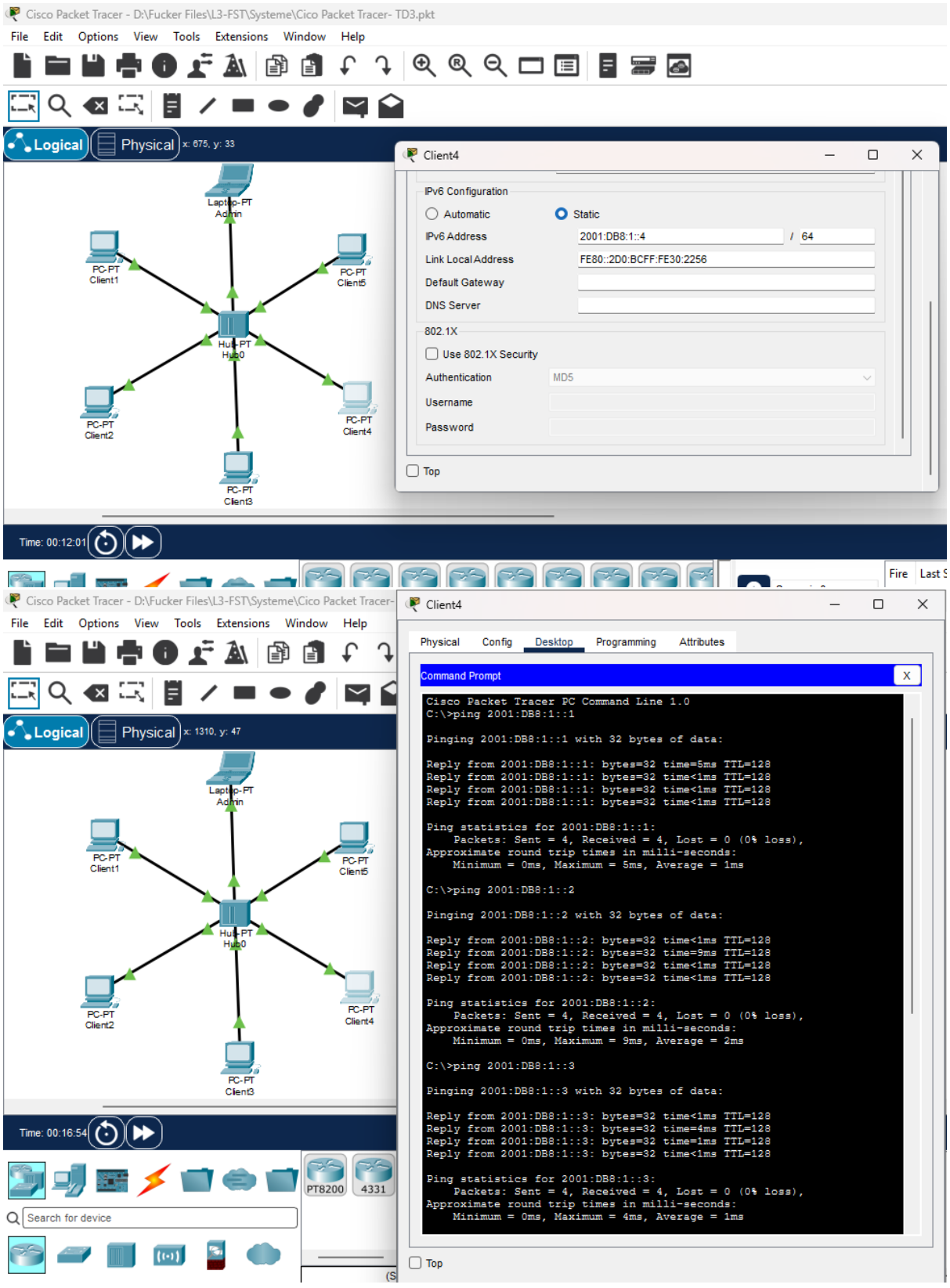
3- Configuration et verification du **Client3** en IPv6 :

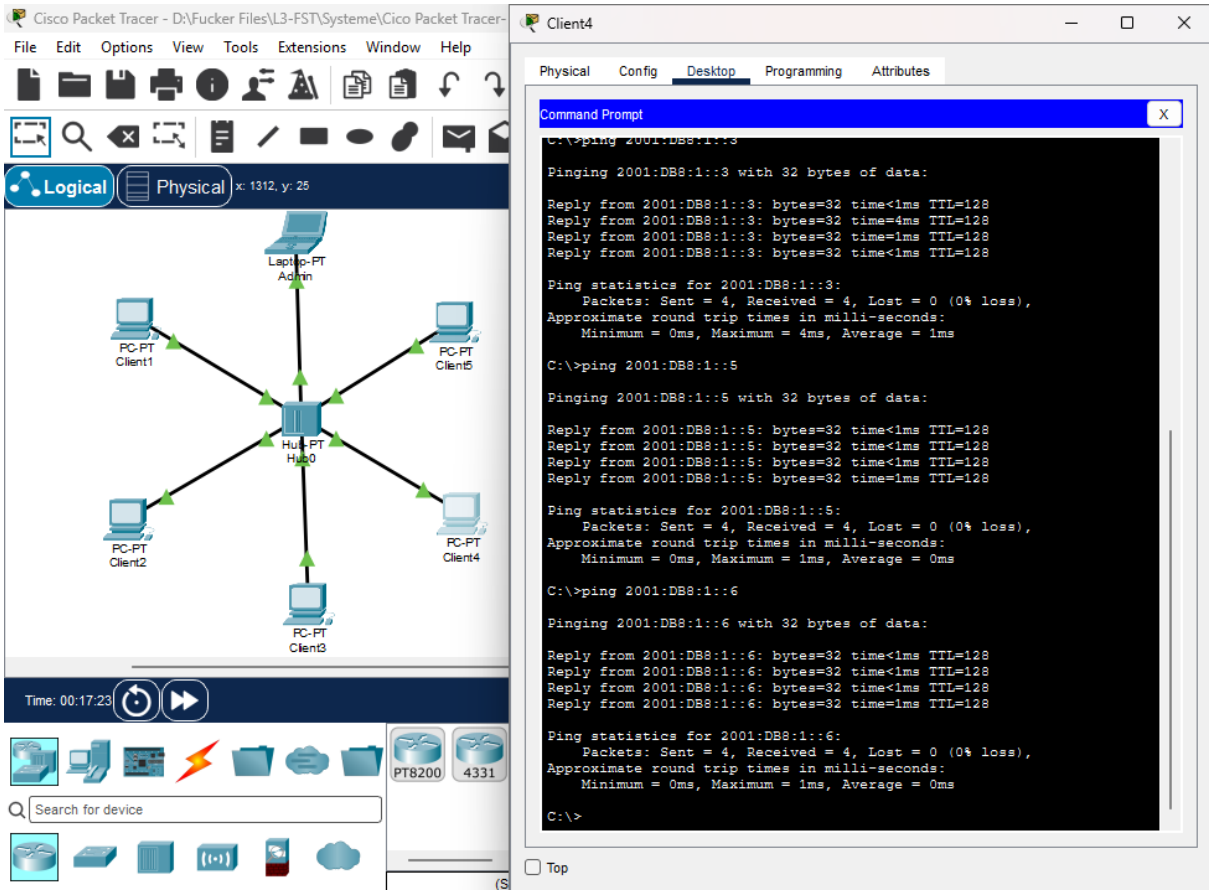

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 673, y: 163
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client3
Automatic Static
IPv6 Address 2001:DB8:1::3 / 64
Link Local Address FE80::202:16FF:FE92:C794
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer-
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1322, y: 30
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:10:36
PT8200
4331
Search for device
Client3
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::2
Pinging 2001:DB8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:DB8:1::4
Pinging 2001:DB8:1::4 with 32 bytes of data:
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top

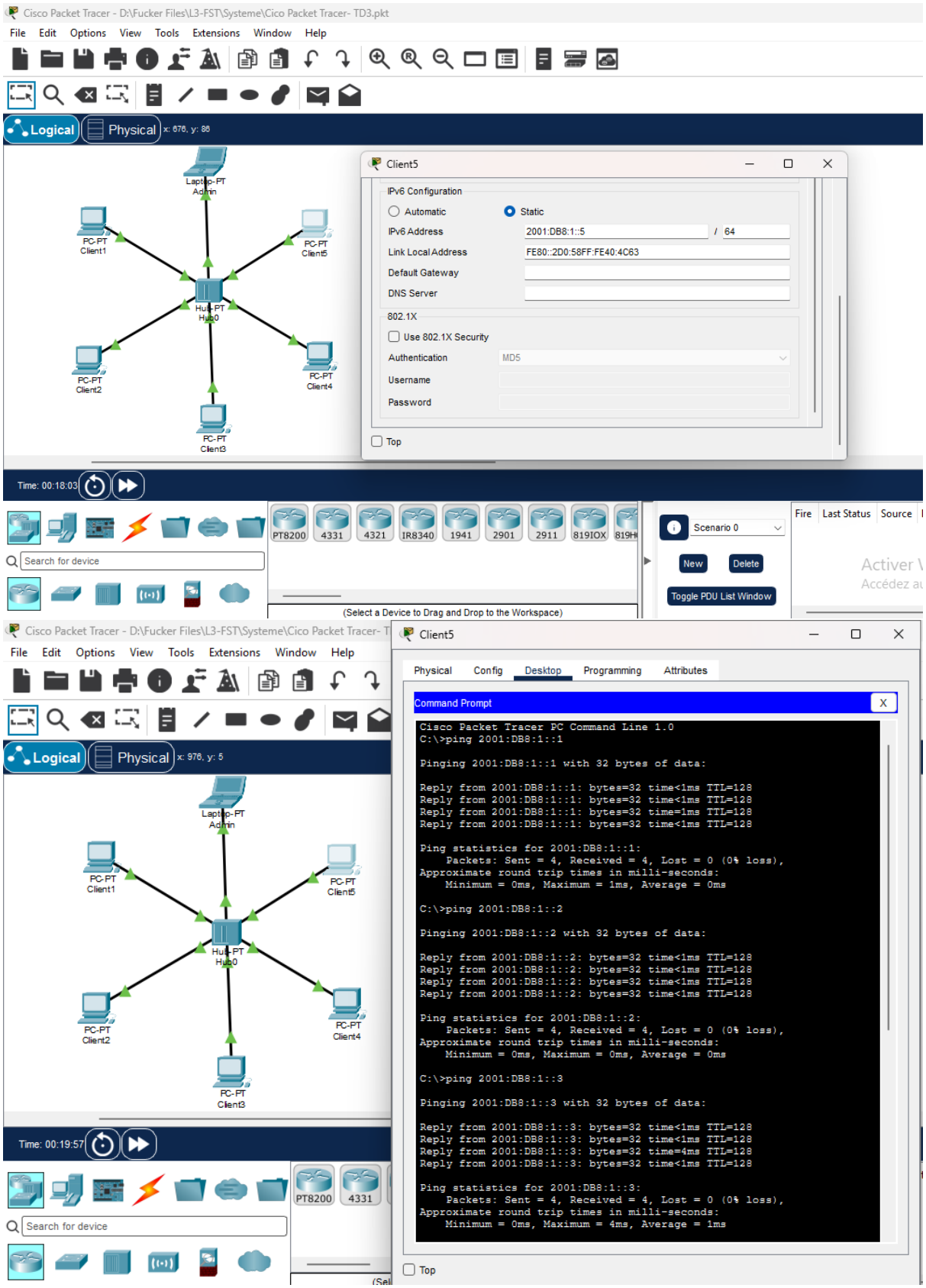


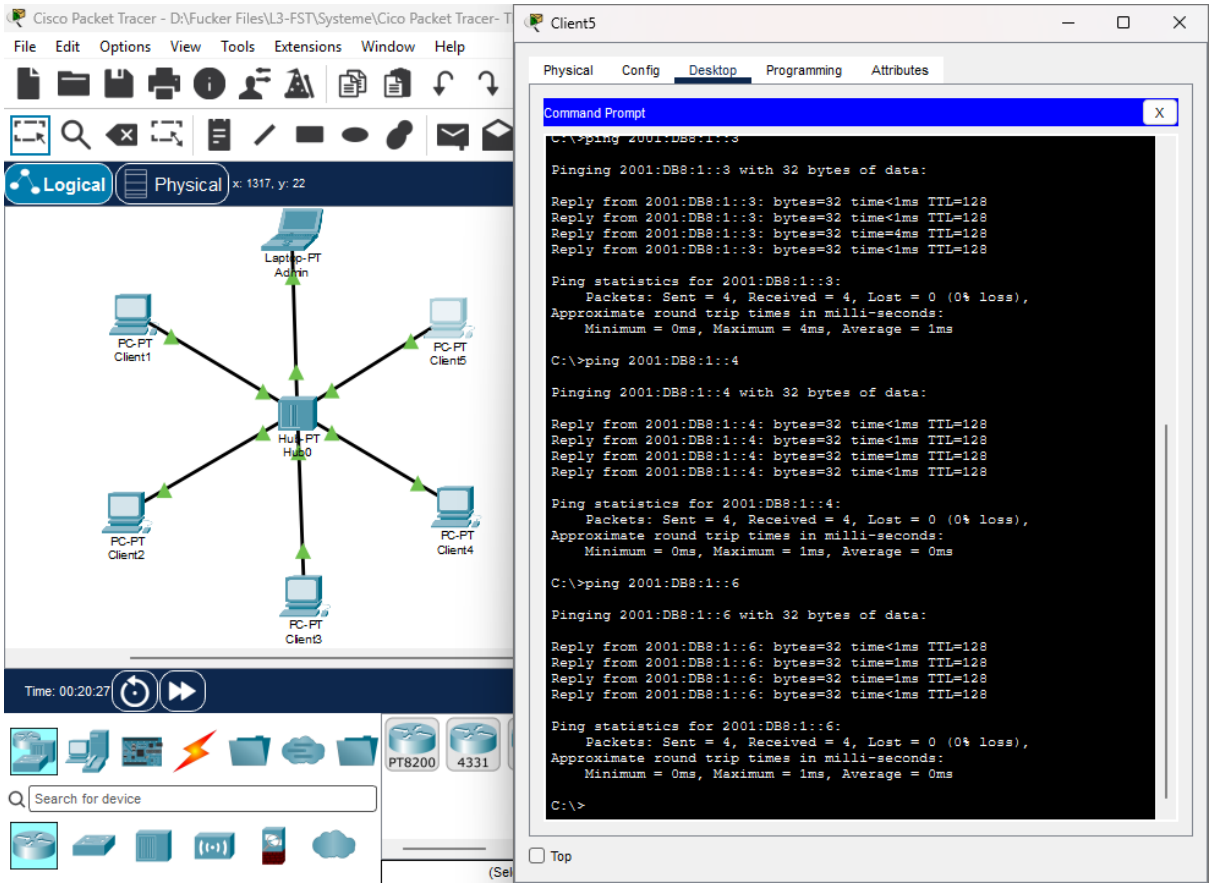
4- Configuration et verification du **Client4** en IPv6 :


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 675, y: 33
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:12:01
Client4
IPv6 Configuration
Automatic
Static
IPv6 Address 2001:DB8:1::4 / 64
Link Local Address FE80::2D0:BCFF:FE30:2256
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer-
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1310, y: 47
Time: 00:16:54
Search for device
PT8200
4331
Client4
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time=5ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 1ms
C:\>ping 2001:DB8:1::2
Pinging 2001:DB8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=9ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 9ms, Average = 2ms
C:\>ping 2001:DB8:1::3
Pinging 2001:DB8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 4ms, Average = 1ms
Top

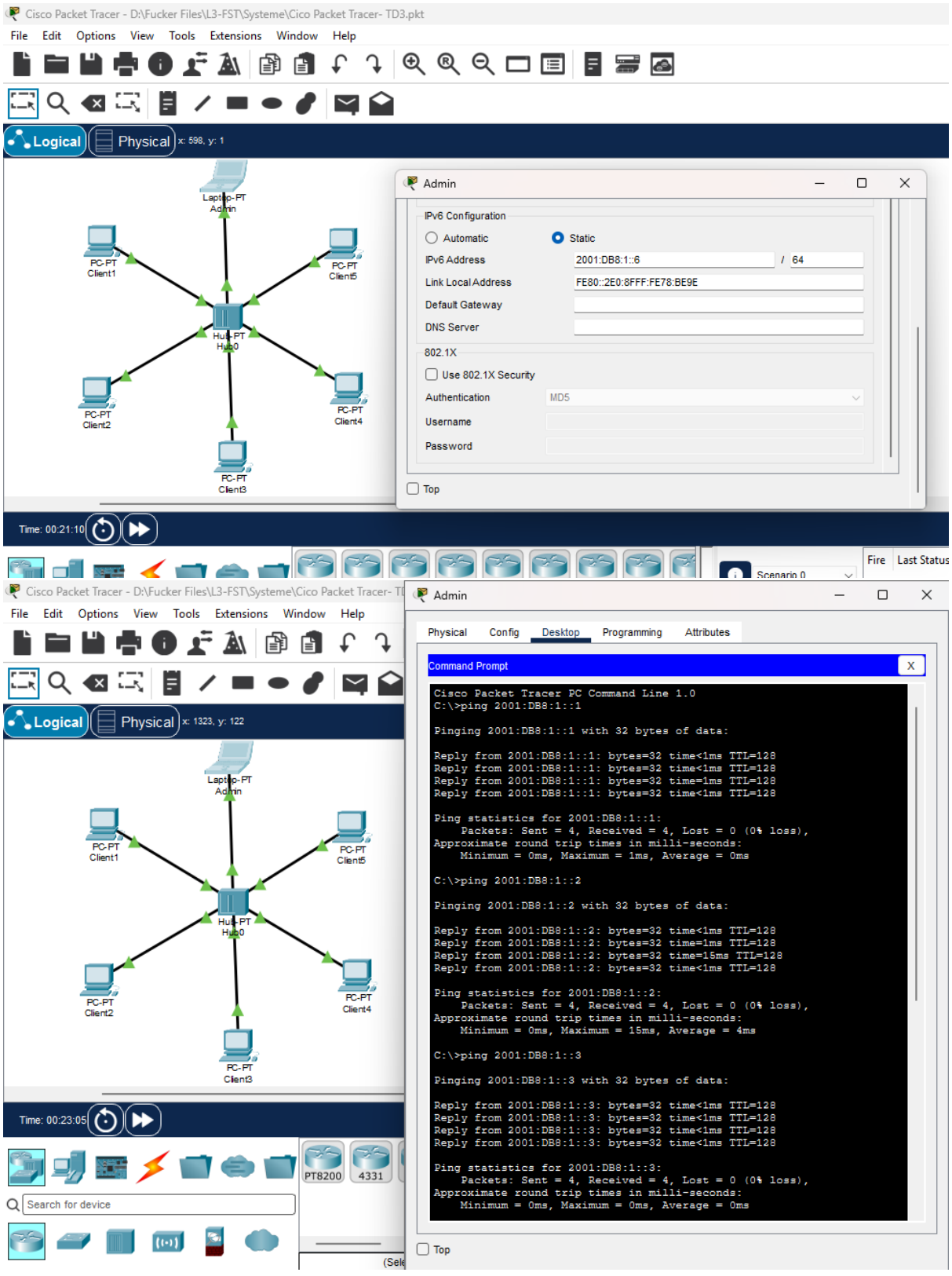


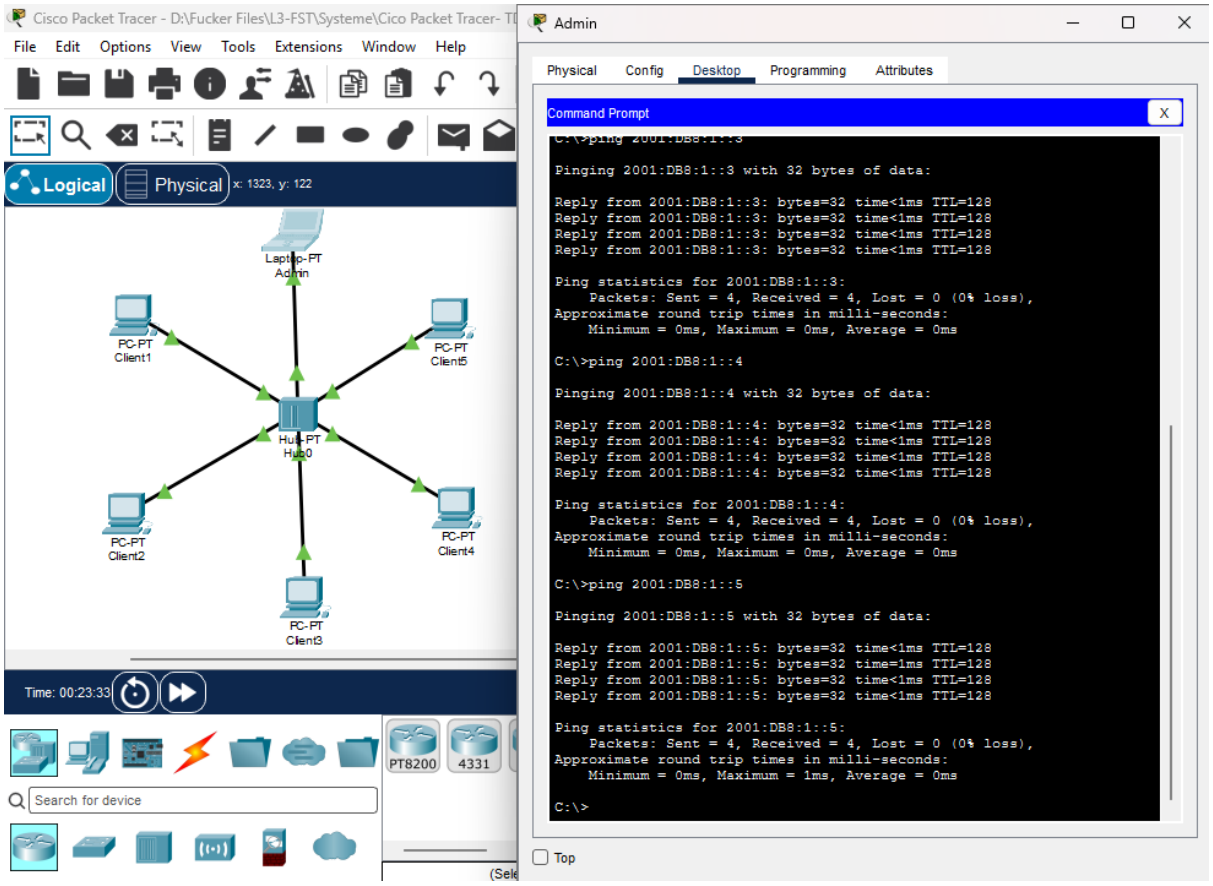
5- Configuration et vérification du **Client5** en IPv6 :


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 676, y: 86
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client5
IPv6 Configuration
Automatic
Static
IPv6 Address 2001:DB8:1::5 / 64
Link Local Address FE80::2D0:58FF:FE40:4C63
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Time: 00:18:03
Search for device
PT8200 4331 4321 IR8340 1941 2901 2911 819IOX
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source
(Select a Device to Drag and Drop to the Workspace)
Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- T
File Edit Options View Tools Extensions Window Help
Logical Physical x: 976, y: 5
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:19:57
Search for device
PT8200 4331
Client5
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::2
Pinging 2001:DB8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:DB8:1::3
Pinging 2001:DB8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 4ms, Average = 1ms
Top



6- Configuration et vérification du **Laptop Admin** en IPv6 :


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 598, y: 1
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Admin
IPv6 Configuration
Automatic
Static
IPv6 Address 2001:DB8:1::6 / 64
Link Local Address FE80::2E0:8FFF:FE78:BE9E
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Time: 00:21:10
Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1323, y: 122
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:23:05
Search for device
PT8200
4331
Admin
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::2
Pinging 2001:DB8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=15ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 15ms, Average = 4ms
C:\>ping 2001:DB8:1::3
Pinging 2001:DB8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top



Dispositifs	IPv4	IPv6
Client1	192.168.1.10	2001:DB8:1::1/64
Client2	192.168.1.11	2001:DB8:1::2/64
Client3	192.168.1.12	2001:DB8:1::3/64
Client4	192.168.1.13	2001:DB8:1::4/64
Client5	192.168.1.14	2001:DB8:1::5/64
Admin	192.168.1.15	2001:DB8:1::6/64

En conclusion, ce travail de laboratoire nous a permis de maîtriser les fondamentaux de l'adressage logique et de la connectivité réseau locale sous Cisco Packet Tracer. La mise en œuvre d'une topologie en étoile à l'aide d'un Hub a démontré avec succès la configuration manuelle et la coexistence des protocoles IPv4 et IPv6 sur des hôtes terminaux. Les tests d'écho systématiques (**ping**) ont affiché un taux de réussite de 100%, validant ainsi la parfaite communication entre l'ensemble des clients et la station d'administration.

Difficultés rencontrées et solutions :

Lors de la configuration IPv6, l'adresse *3000:is9:1::X* fournie dans le document du TD3 a été rejetée par Cisco Packet Tracer. En faisant des recherches , nous avons constaté que l'adressage IPv6 utilise exclusivement une notation hexadécimale (0-9, A-F). La présence de la lettre « s » rendait donc l'adresse invalide. Pour résoudre ce problème et mener à bien le TD, nous avons adapté le plan d'adressage en remplaçant la syntaxe incorrecte par le réseau valide *2001:DB8:1::X/64*.